

Task 1: Basic Aggregate Functions

You are given a table:

Employees(EmployeeID, Name, DepartmentID, Salary)

Write SQL queries to:

1. Find the total salary of all employees
2. Find the average salary
3. Find the highest salary
4. Find the lowest salary
5. Count the total number of employees

Requirements

- Use aggregate functions: SUM(), AVG(), MAX(), MIN(), COUNT()
 - Do not use GROUP BY in this task
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Task 2: GROUP BY with Departments

Using the same **Employees** table:

Write SQL queries to:

1. Show total salary per department
2. Show average salary per department
3. Show number of employees in each department

Requirements

- Use GROUP BY DepartmentID
 - Use aggregate functions
 - Order results by DepartmentID
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Task 3: HAVING Clause Filtering

Using the **Employees** table:

Write SQL queries to:

1. Show departments where total salary is greater than 10000
2. Show departments with more than 5 employees
3. Show departments where average salary is above 5000

Requirements

- Use GROUP BY
 - Use HAVING (not WHERE for aggregate conditions)
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Task 4: Orders Analysis

You are given:

Orders(OrderID, CustomerID, Amount)

Write SQL queries to:

1. Find total revenue from all orders
 2. Find average order amount
 3. Count total orders
 4. Find maximum and minimum order amount
 5. Show total amount per customer
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Task 5: Customers Order Summary

Using:

Customers(CustomerID, CustomerName)

Orders(OrderID, CustomerID, Amount)

Write SQL queries to:

1. Show number of orders per customer
2. Show total amount spent per customer
3. Show customers who spent more than 1000 total

4. Show customers with no orders

Requirements

- Use JOIN + aggregate functions
 - Use GROUP BY CustomerID
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Task 6: Sales Performance

You are given:

Sales(SaleID, ProductID, Quantity, Price)

Write SQL queries to:

1. Find total sales revenue (Quantity × Price)
 2. Find total quantity sold per product
 3. Find average price per product
 4. Find best-selling product (highest quantity sold)
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Task 7: Mixed Aggregation Challenge

Using:

OrderDetails(OrderID, ProductID, Quantity)

Products(ProductID, ProductName)

Write SQL queries to:

1. Show total quantity sold per product name
2. Show products with total quantity greater than 50
3. Show number of orders per product
4. Find the most ordered product

Task 8: Department Salary Statistics

Table:

Employees(EmployeeID, Name, DepartmentID, Salary)

Write SQL queries to:

1. Show maximum salary per department
2. Show minimum salary per department
3. Show average salary per department
4. Show salary difference (MAX - MIN) per department

Requirements

- Use GROUP BY DepartmentID
 - Use multiple aggregate functions in one query
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Task 9: Employee Performance Analysis

Table:

Employees(EmployeeID, Name, DepartmentID, Salary)

Write SQL queries to:

1. Show departments where average salary is above 6000
2. Show departments where total salary is above 20000
3. Show departments with more than 3 employees
4. Sort results by highest average salary

Requirements

- Use HAVING
 - Use ORDER BY
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Task 10: Customer Spending Report

Tables:

Customers(CustomerID, CustomerName)

Orders(OrderID, CustomerID, Amount)

Write SQL queries to:

1. Show total spending per customer
 2. Show average order value per customer
 3. Show number of orders per customer
 4. Show only customers who spent more than 500
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Task 11: Product Sales Summary

Table:

Sales(SaleID, ProductID, Quantity, Price)

Write SQL queries to:

1. Calculate total revenue per product
2. Find total quantity sold per product
3. Find average price per product
4. Show products where revenue is greater than 1000

Hint:

Revenue = Quantity × Price

Task 12: Order Analysis with Aggregation

Table:

Orders(OrderID, CustomerID, Amount)

Write SQL queries to:

1. Find total number of orders
2. Find total revenue

3. Find average order amount
4. Find highest and lowest order amount
5. Show customers with more than 2 orders

Task 13: Group + Join Challenge

Tables:

Orders(OrderID, CustomerID, Amount)

Customers(CustomerID, CustomerName)

Write SQL queries to:

1. Show total orders per customer name
2. Show total spending per customer name
3. Show only customers with spending above 1000
4. Sort by highest spending

Task 14: Advanced HAVING Practice

Table:

Employees(EmployeeID, Name, DepartmentID, Salary)

Write SQL queries to:

1. Show departments where MAX salary > 8000
2. Show departments where MIN salary < 3000
3. Show departments where AVG salary between 4000 and 7000
4. Show departments with at least 2 employees

Task 15: Real Business Scenario

Tables:

Sales(SaleID, ProductID, Quantity, Price)

Products(ProductID, ProductName)

Write SQL queries to:

1. Show total revenue per product name
2. Show best-selling product (by quantity)
3. Show most profitable product (by revenue)
4. Show products with no sales (if any exist)

Task 16: Multi-Level Aggregation

Table:

Employees(EmployeeID, Name, DepartmentID, Salary)

Write SQL queries to:

1. Show total salary per department
2. Show average salary per department
3. Show departments where total salary is above the overall average department salary
4. Rank departments by total salary (conceptually)

Requirements

- Use GROUP BY
- Use subquery with aggregate function

Task 17: Sales Time Analysis

Table:

Sales(SaleID, ProductID, Quantity, Price, SaleDate)

Write SQL queries to:

1. Show total revenue per day
2. Show total revenue per month
3. Find the day with highest sales revenue
4. Count number of sales per month

Requirements

- Use GROUP BY SaleDate (or DATE functions)
- Use SUM, COUNT

Task 18: Customer Loyalty Analysis

Tables:

Customers(CustomerID, CustomerName)

Orders(OrderID, CustomerID, Amount)

Write SQL queries to:

1. Show customers with more than 5 orders
2. Show customers whose average order value is above 200
3. Show top 3 customers by total spending
4. Find customers who placed only one order

Requirements

- Use GROUP BY
 - Use HAVING
 - Use ORDER BY + LIMIT
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Task 19: Product Category Performance

Tables:

Products(ProductID, ProductName, CategoryID)

Sales(SaleID, ProductID, Quantity, Price)

Categories(CategoryID, CategoryName)

Write SQL queries to:

1. Show total revenue per category
2. Show average product revenue per category

3. Show categories with revenue > 5000
4. Show best category by total sales

Task 20: Null Handling in Aggregation

Table:

Employees(EmployeeID, Name, DepartmentID, Salary)

Write SQL queries to:

1. Count employees with non-null salary
2. Show departments including those with NULL salaries
3. Calculate average salary ignoring NULL values
4. Show departments where all salaries are NULL